1 **(a)** 3.068822... **[B2]**

If 0 scored, allow [B1] for seeing 15.36 or 2.485... or 0.81

(b) 3.069 [B1] (strict FT from their (a) if given to at least 4 d.p.)

[Total Marks: 3]

2 **(a)** Numerator $\sqrt{22.01} = 4.69148\ldots$ or Denominator $12.167 - 1.4 = 10.767$ seen **[M1]**

0.435727839 **[A1]**

(b) 0.436 **[B1]**

[Total Marks: 3]

3 **(a)** 8.40 **[B1]**

(b) 5 hours 9 minutes **[B1]**

(c) $280 \div 85$ **[M1]**

3.294... **[A1]**

3 hours 18 minutes **[A1]**

[Total Marks: 5]

4 **(a)** Time $= 3 + \dfrac{40}{60}$ or $\dfrac{11}{3}$ hours **[M1]**

Cost $= 45 + 32.50 \times \dfrac{11}{3}$ **[M1]**

164.17 **[A1]**

(b) $230.25 - 45 = 185.25$ **[M1]**

$\dfrac{185.25}{32.50} = 5.7$ hours **[M1]**

5 hours 42 minutes **[A1]**

[Total Marks: 6]

5 **(a)** $\dfrac{4}{3}\pi(1.4)^3$ **[M1]**

11.49404... **[A1]**

(b) their 11.494... $\div 0.035$ **[M1]**

328.4... or 328 (minutes) **[A1]**

their 328.4... $\div 60$ (converting to hours) **[M1]**

5 hours 28 minutes **[A1]**

[Total Marks: 6]

6 **(a)** $\frac{1}{3} \times \pi \times 3.2^2 \times (3 \times 8.45 - 3.2)$ **[M1]**

237.5 **[A1]**

(b) $\frac{3V}{\pi h^2} = 3R - h$ **[M1]**

$\frac{3V}{\pi h^2} + h = 3R$ **[M1]**

$R = \frac{V}{\pi h^2} + \frac{h}{3}$ or $R = \frac{3V + \pi h^3}{3\pi h^2}$ **[A1]**

(c) $\frac{500}{\pi \times 5.6^2} + \frac{5.6}{3}$ **[M1]**

6.94 **[A1]**

[Total Marks: 7]